



Improvement of Phase Change Materials in Thermal Energy Storage Systems: A Comprehensive Review

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Abstract: Phase change materials (PCMs) are highly effective in storing and releasing thermal energy during phase transitions, making them critical for thermal energy storage (TES) systems, particularly for renewable energy sources such as solar and wind. They have been reported in energy storage, especially in renewable energy systems such as solar and wind. However, despite their potential, their practical use is limited by low thermal conductivity and slow heat transfer rates. These limitations reduce the efficiency of PCMs in applications requiring rapid thermal responses, such as solar thermal storage and electric vehicle (EV) battery cooling. This review synthesizes and compares recent numerical and experimental studies on PCM enhancement techniques. A significant challenge across these studies is the lack of uniform operating conditions, which complicates the identification of the most effective methods for specific TES applications. The review highlights several strategies to improve PCM performance, including the use of metal foams (MFs), nanoparticles (NPs), and fins. MF has been shown to significantly improve thermal conductivity, increasing it by up to 200% for calcium chloride hexahydrate and 100% for paraffin, while also reducing melting times by 84.9% compared to pure paraffin. NPs, like copper oxide (CuO) and aluminum oxide (Al₂O₃), can enhance thermal conductivity by up to 122% relative to pure PCM. However, higher concentrations of NPs may increase viscosity, which slightly hinders heat transfer. Fins provide a cost-effective method to enhance heat transfer. The addition of fins has been shown to reduce melting times by 65.5% at 3600 seconds, making them an ideal choice for applications where cost is a key consideration. Hybrid systems combining MFs and NPs achieve the greatest performance improvements. For instance, using 3% NPs and a 60% porosity in copper MF increases thermal conductivity by 37.7% and reduces the melting time by 87.03%. Further improvements are observed when using MF with 85–90% porosity and 10–15% NPs, achieving a 90% reduction in melting time. This demonstrates the synergistic effect of combining these two techniques. In conclusion, hybrid methods combining MFs and NPs offer an efficient and cost-effective approach for enhancing PCM performance in TES applications. By integrating the strengths of these techniques, multiple performance limitations can be addressed simultaneously, providing a viable solution for large-scale TES systems.

Keywords: Phase change materials; Metal foam; Nanoparticle; Fin; Thermal energy storage; Heat transfer enhancement; Hybrid system

1 Introduction

Thermal energy storage (TES) systems are essential for efficiently managing intermittent renewable energy sources such as solar and wind power. These systems store excess energy for later use, enhancing the stability of power grids and mitigating the variability of renewable sources. Phase change materials (PCMs) are especially valuable in TES systems due to their ability to store energy as latent heat. During phase transitions between the solid and liquid states, PCMs absorb and release substantial amounts of heat, which makes them highly efficient for temperature regulation and energy storage [1, 2]. This property enables them to optimize the performance of heating and cooling systems, reducing reliance on traditional heating, ventilation, and air conditioning (HVAC) systems [3–5]. However, despite their advantages, the widespread adoption of PCMs is limited by their low thermal conductivity. This limitation significantly impairs their heat-transfer efficiency, which is crucial for applications that require rapid thermal responses. While PCMs excel in energy storage, their slow heat-transfer rates render them

less suitable for applications requiring rapid temperature changes, such as solar thermal storage and electric-vehicle battery cooling [3, 6, 7]. As a result, there has been a significant research effort to overcome these challenges and improve PCM performance across various applications. Recent studies have focused on several strategies to enhance PCM performance, particularly to address thermal conductivity. For instance, Kiyak et al. [8] investigated the impact of heater location (bottom, side, and top) and enclosure shape (square and circular) on the thermal performance of PCM systems. Their finding suggested that optimizing the placement of the heat source and the shape of the enclosure could substantially improve heat transfer rates, which is crucial for applications requiring rapid absorption and release of thermal energy. Other studies have explored innovative PCM system configurations to improve heat transfer. Nithyanandam and Pitchumani [9] demonstrated that cascaded PCM systems, where layers of different PCMs are strategically stacked, can enhance heat transfer and minimize temperature fluctuations. This configuration enables more efficient thermal management by improving the system's overall thermal conductivity. Further research by Jaworski et al. [10] and Souayfane et al. [4] emphasized the importance of combining numerical simulations and experimental approaches in evaluating PCM systems, reporting energy reductions of up to 40% in certain systems. Particularly with respect to improving the thermal conductivity of PCMs and ensuring long-term durability.

Researchers, including Cabeza et al. [11] and Osterman et al. [12], have continued to investigate various methods to enhance PCM performance. Notable strategies include encapsulation and the use of composite materials, which have shown promise in addressing the inherent limitations of PCMs. Encapsulation in particular allows for better containment of the PCM, thereby improving its thermal stability and prolonging its lifecycle. In recent years, the integration of composite materials, metal foam (MFs), and nanoparticles (NPs) has emerged as a significant advancement. This approach not only enhances the thermal conductivity of PCMs but also increases their energy storage capacity and shortens their phase-change times. Such improvements are crucial for making PCMs more suitable for large-scale TES applications, where efficiency and rapid thermal response are paramount [7, 13]. By enhancing both the thermal properties and energy storage performance, these combined strategies have the potential to significantly improve the viability of PCMs in real-world, large-scale TES applications.

2 Classification of Phase Change Materials

PCMs are categorized into two types based on the phase change they undergo (e.g., solid-to-liquid [7]) and on their chemical composition (organic, inorganic, and composite materials [14]), as illustrated in Figure 1. Each category has unique advantages and limitations depending on the application.

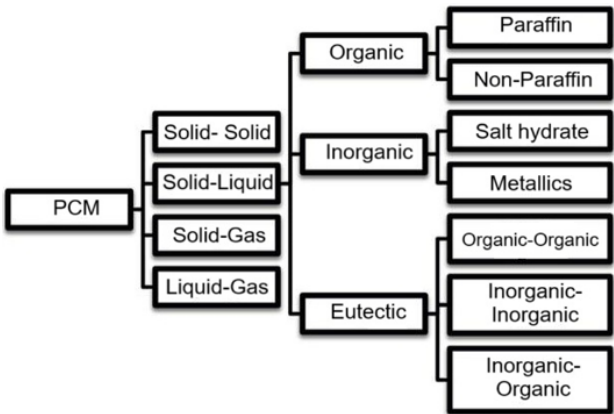


Figure 1. Classification of phase change material (PCM) based on their thermal storage capacities, phase transition temperature, thermal properties, and material type, adapted from Ref. [15] under the Creative Commons CC BY 4.0 license

2.1 Phase Change Materials Classification by Phase Change

Solid to solid: Materials change their internal crystal structure without melting, while still storing or releasing latent heat.

Solid to liquid: During a phase change, thermal energy is absorbed or released, and the temperature remains constant as materials transition between solid and liquid states. Solid-to-liquid PCMs are the most commonly used in TES systems due to their efficient storage and release of thermal energy.

Solid to gas: The direct change from solid to gas, called sublimation, occurs when a solid absorbs enough energy to skip the liquid phase and go straight into the gas phase. This process is rarely used because it requires a very high level of energy and is difficult to control.

Liquid to gas: The change from liquid to gas, called vaporization, happens when a liquid absorbs enough energy for its molecules to break free from intermolecular forces and turn into vapor [16].

2.2 Phase Change Materials Classification by Composition

Organic materials, such as paraffin and fatty acids, are commonly used for TES due to their high latent heat of fusion, chemical stability, and low corrosiveness; they are ideal for heat storage. However, they have low thermal conductivity, which limits their applications in systems that require rapid thermal response, such as solar thermal storage [11, 17].

Inorganic materials, such as salt hydrates and metal alloys, have higher thermal conductivity than organic materials, making them more suitable for heat transfer applications. However, these materials face challenges, including supercooling and corrosion, which can reduce their long-term efficiency in TES systems [6].

Composite materials combine the advantages of both organic and inorganic materials or incorporate conductive materials such as metals or carbon nanotubes. This combination enhances their overall thermal performance, making them more efficient for both energy storage and heat transfer [4].

3 Applications for Phase Change Materials

PCMs are integrated across various industries, including construction, solar energy, EVs, electronics, and smart textiles, as illustrated in Figure 2, which shows their integration in each application due to their efficient TES and release. These materials help regulate building temperatures, manage energy in solar power systems, and enhance the performance of electric vehicle (EV) batteries. In the construction industry, PCMs are integrated into both passive and active energy systems to improve energy efficiency and reduce reliance on traditional heating, ventilation, and air conditioning (HVAC) systems. In passive systems [3], PCMs absorb heat during the day and release it at night, stabilizing indoor temperature and reducing energy consumption. In active systems, PCMs are paired with mechanical devices [18], such as fans, to accelerate heat transfer, leading to significant energy savings [11, 12]. PCMs are also crucial in solar energy systems by storing excess energy during the day and releasing it at night or on cloudy days [9, 13, 19]. This ensures a continuous energy supply and enhances the reliability of solar power systems, particularly in regions with intermittent sunlight [6, 20]. In EVs and aerospace applications, PCMs regulate battery and equipment temperatures, preventing overheating and ensuring optimal performance. The integration of PCMs into battery thermal management systems ensures rapid heat dissipation, which is crucial for battery life and efficiency [21]. In electronics, PCMs help manage heat in devices such as smartphones and laptops, ensuring they remain within their optimal operating temperature range under heavy use [22, 23]. PCMs are increasingly integrated into smart textiles to regulate body temperature. These textiles absorb heat when the body is warm and release it when the body cools down, improving comfort and performance in clothing for athletes and outdoor enthusiasts. This technology has been applied to athletic wear and outdoor clothing [24]. In the food industry, PCMs are incorporated into packaging materials to maintain the temperature of perishable goods during transportation. This technology ensures product quality and safety by providing temperature control throughout the supply chain [25].

4 Enhancement of Phase Change Materials Performance

This section analyzes various studies on enhancing PCMs by focusing on heat-transfer mechanisms, key geometric parameters, system scale, and operating conditions, particularly in comparison with pure-PCM systems. The goal of this analysis is to understand the impact of these factors on the thermal performance of PCM systems and to evaluate strategies to overcome inherent limitations, such as low thermal conductivity and slow heat transfer rates. In the PCM system, heat transfer primarily occurs through conduction, convection, and radiation. The role of geometric parameters such as porosity, fin size, and their arrangement has been reported to significantly influence heat transfer rates. For instance, materials with high porosity, such as MF, can affect thermal conductivity, potentially enhancing heat-transfer properties. Moreover, the design and strategic placement of fins increase the surface area, thereby facilitating convective heat transfer and improving overall heat transfer efficiency. As PCM systems are scaled up from laboratory prototypes to real-world applications, several challenges emerge, particularly concerning heat dissipation, PCM volume, and overall system design. Larger systems often struggle to ensure uniform heat distribution and control temperature gradients, which can undermine both thermal storage capacity and system efficiency. The scalability of PCM systems is therefore closely linked to addressing these issues at scale. Operational conditions, including temperature ranges, pressure, and the nature of heat sources, play a crucial role in determining the behavior and effectiveness of PCMs. Higher operating temperatures can accelerate the phase change process, but they also introduce the risk of material degradation over time. Additionally, the nature of heat sources, whether solar radiation, waste heat from batteries, or other sources affect the efficiency with which PCMs absorb and release thermal energy. Variations in operational conditions between laboratory and industrial settings can significantly influence the applicability of laboratory results to real-world scenarios. Factors such as temperature fluctuations and changes in material properties under varying conditions can alter the performance of PCM-enhanced systems outside

controlled environments. A number of techniques have been proposed to enhance the heat-transfer performance of PCMs. This includes integrating MFs, fins, and NPs, each of which has been shown to improve thermal conductivity and reduce phase-change duration, as shown in Figure 3.

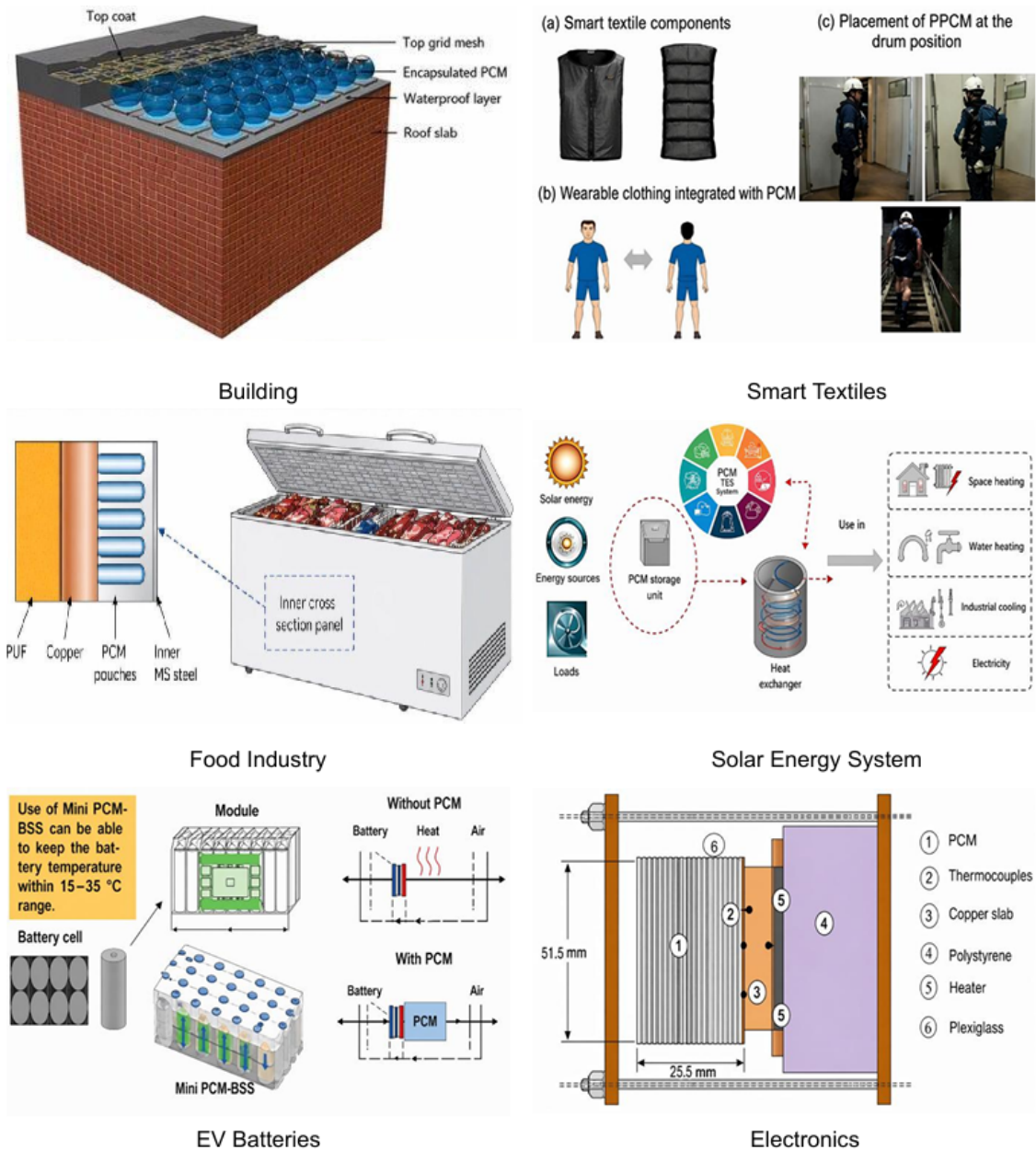


Figure 2. Applications of phase change materials (PCMs) in various energy systems, including thermal energy storage (TES) for building, smart textiles, solar energy systems, the food industry, battery thermal management, and electronics

Note: EV = electric vehicle. Building, adapted with permission from Ref. [26] © 2025 Elsevier; Smart Textiles, adapted from Ref. [27] under the Creative Commons CC BY 4.0 license; Food Industry, adapted from Ref. [28] under the Creative Commons CC BY 4.0 license; Solar Energy System, adapted with permission from Ref. [29] © 2025 Elsevier; EV Batteries, adapted from Ref. [30] under the Creative Commons CC BY 4.0 license; Electronics, adapted with permission from Ref. [31] © 2015 Elsevier.

For example, incorporating high-thermal-conductivity materials, such as MFs, into PCM systems has been shown to significantly improve heat transfer, benefiting applications requiring rapid thermal response. However, the introduction of MFs can also reduce the PCM volume, potentially compromising the system's overall energy storage capacity. Thus, a delicate balance must be struck between improving thermal performance and maintaining storage efficiency. NPs have also been found to improve heat transfer by stabilizing the PCM, preventing phase separation during repeated melting and solidification cycles. The addition of NPs creates additional thermal pathways, thereby increasing the effective surface area of the PCM and enhancing thermal performance. However, the concentration of

NPs needs to be carefully optimized, as high concentrations can increase the viscosity and slow natural convection, potentially diminishing the overall heat transfer efficiency. Therefore, it is critical to find an optimal NP concentration that maximizes thermal conductivity benefits without introducing viscosity-induced limitations.

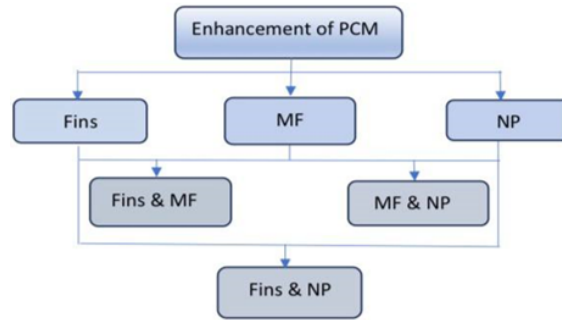


Figure 3. Enhancement of phase change material (PCM) using various methods, such as metal foams (MFs), nanoparticles (NPs), and fins

4.1 Integration with Metal Foam

A significant body of research has focused on incorporating high-thermal-conductivity materials, such as MFs, into PCM systems to enhance heat transfer performance. The studies reviewed consistently evaluate improvements by comparing against reference cases using pure PCM, under identical geometry and boundary conditions. It is essential to clarify these reference conditions to ensure comparability of results across studies. Incorporating a high-thermal-conductivity matrix, such as aluminum foam, has been reported to significantly accelerate melting rates. A numerical analysis of conduction-dominated heat transfer in PCM storage systems has demonstrated that the optimal foam porosity is approximately 95%, as it balances enhanced heat transfer with sufficient PCM volume retention. This analysis emphasizes the critical role of porosity in heat-transfer performance under constant-wall-temperature boundary conditions [32].

Further numerical studies have reported that even a small amount of MF can have a profound effect on the melting behavior. For example, adding copper MF reduced the melting time by 80%, with the analysis considering both conduction and natural convection. The study also examined the influence of foam configuration on thermal performance and assessed the economic implications of incorporating MF under constant wall-temperature conditions [33]. In addition to numerical simulations, experimental studies have also highlighted the benefits of MF integration. One such study demonstrated that incorporating MF into heat sinks substantially improved thermal conductivity, resulting in a noticeable enhancement in heat dissipation compared to a traditional heat sink [34]. Similarly, an experimental investigation involving MF blocks in a horizontal channel reported a significant increase in heat transfer efficiency, outperforming the baseline channel without foam [35]. Nima and Ali [36] conducted a numerical analysis of the effect of MF blocks on heat transfer in a flat-plate solar collector. The study revealed a considerable increase in thermal efficiency, with notable improvements in heat absorption and distribution compared to standard collectors. Meanwhile, Ali and Ghashim [37] further investigated the thermal performance of MF in pipes, demonstrating a marked improvement in heat dissipation over conventional pipe designs and a substantial increase in thermal efficiency.

Another experimental study explored the role of copper MF with 97.3% porosity in a phase change composite material consisting of paraffin impregnated with copper foam. The study reported a 40–50% reduction in melting time and a significant enhancement in heat conduction within the PCM. The foam's porous structure minimized the effects of natural convection, highlighting the importance of foam structure in optimizing thermal performance under controlled boundary conditions [38]. The effect of MF porosity on heat transfer was also explored in several studies. One investigation found that higher porosity (96.6%) accelerated melting through convection, while lower porosity (88.4%) led to more uniform heat distribution via conduction, though with a smaller effect on melting time. This study demonstrated that reducing porosity and pore size improved heat conduction within the PCM, speeding up melting under controlled heating conditions [39]. A porosity of 93.9% was found to offer the best thermal performance for PCM-embedded MF systems. In line with these findings, numerical studies of phase-change heat transfer in an enclosure with uniform and heterogeneous MF layers showed that a modest 7.5% increase in porosity could lead to a 66% reduction in melting time. The analysis considers both conduction and natural convection, using a neural-network approach to optimize foam-layer configurations to improve thermal performance under specific boundary conditions [40]. These results are visually represented in Figure 4. Li et al. [41] reported that copper foam with 97% porosity and 25 pore per inch (PPI) increased thermal conductivity by 69.1%, further highlighting the substantial impact of MF properties on thermal performance of PCM systems. To enhance the melting process and improve

energy storage efficiency, combined experimental and numerical investigations were conducted on the melting of paraffin-filled open-cell metallic foams. The presence of MF significantly boosted heat transfer by increasing the effective thermal conductivity. However, the foam's porous structure also suppressed natural convection, a factor that must be carefully considered. The incorporation of MF into the PCM system resulted in a tenfold increase in heat transfer, drastically reducing both melting and freezing times. The size of the MF is a critical factor influencing heat transfer performance.

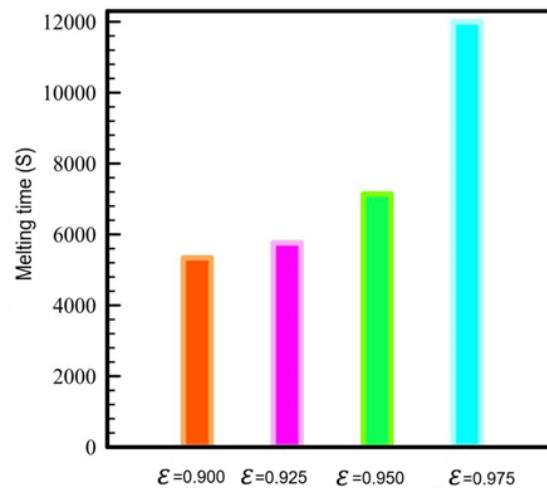


Figure 4. The melting time when the melting volume fraction (MVF) reaches 0.95 for different porosity values, adapted with permission from Ref. [40] © 2024 Elsevier

Numerical studies indicate that when the MF matrix exceeds 15% of the total system volume, the improvements in heat transfer begin to decline. Therefore, the optimal foam size for maximizing heat transfer lies between 5% and 15% of the total system volume. This conclusion was derived from numerical analysis of heat transfer enhancement in PCM by embedding a metallic structure in the storage area, with a primary focus on conduction-dominated heat transfer. The findings showed that the inclusion of MF in the PCM accelerated the melting rate under steady thermal boundary conditions [42]. Thus, the melting time is not solely determined by porosity; factors such as pore geometry and the specific location of the MF also significantly influence heat transfer performance. Further, investigations examined the effect of copper foam microstructure on heat transfer within PCM systems. Numerical analysis revealed that altering the geometry of the copper metal led to 29.9% reduction in melting time, emphasizing the importance of microstructural parameters in enhancing thermal conductivity during the phase change. Additionally, the placement of the MF within the PCM system proved to be crucial for optimizing heat transfer efficiency [43]. Yang et al. [44] and Xu et al. [33] demonstrated that strategically positioning MF within both the PCM and the Heat Transfer Fluid (HTF) significantly enhances heat transfer performance. Xu et al. [33] found that relocating the heat source from the bottom to the side of the system resulted in a 70.5% reduction in melting time, whereas positioning it at the top yielded only a modest 4.7% improvement. Yang et al. [44] further explored the effect of a porous MF on heat-transfer enhancement in a thermal-energy-storage tube using both numerical and experimental methods. Their study reported that the interaction between foam structure and natural convection plays a key role in improving melting performance. Chen et al. [45] investigated the optimal configuration for a PCM-based TES unit enhanced with MF. They found that the best performance occurred when the MF was placed in both the PCM and the HTF, with a pore density of 15 PPI and under higher heat input conditions (80 W). Their study, conducted using numerical analysis, evaluated the effects of conduction and natural convection on thermal performance under controlled laboratory conditions [40]. The angle at which the MF was installed also influenced performance: a 0° angle for foam placement enhanced horizontal heat transfer, while a porosity of 93.9% offered the optimal balance between thermal conductivity and PCM retention. Experimental analysis revealed that a pore size of 30 PPI and 90% porosity were optimal for enhancing phase-change behavior in paraffin wax embedded in copper foams [46].

Numerical studies of a multiple-segment MF–PCM latent heat storage unit further evaluated the effects of porosity, pore density, and heat source location. The analysis emphasized the role of foam segmentation in improving melting performance. The findings highlighted that heat transfer in these systems was primarily driven by conduction, and foam segmentation played a key role in optimizing the melting process [47]. In addition, Almajali et al. [48] investigated the effect of copper-coated carbon foam on heat transfer. Their findings showed that copper coating increased thermal conductivity by a factor of 1.67 relative to untreated carbon foam, resulting in improved heat transfer and reduced melting times. The cost-performance balance is a critical factor when integrating MF into PCM

systems. Experimental investigations, such as those on copper-coated PCM-infiltrated MF composites, have shown that surface coatings can significantly enhance the effective thermal conductivity, thereby improving melting behavior under constant thermal loading. For instance, Krishnan et al. [49] demonstrated that increasing the Nusselt number (Nu) above 5.9 accelerated heat transfer between the foam and the PCM, thereby enhancing thermal efficiency. This was achieved by developing a two-temperature numerical model that describes the solid–liquid phase change in MFs. The model incorporated thermal non-equilibrium between the foam and PCM phases and focused on conduction-dominated melting processes, highlighting the importance of thermal interactions in optimizing PCM performance. Further contributions by Zhu et al. [50] revealed that partially filling the MF with two-thirds of its volume reduced both material costs and weight by 33% without compromising thermal performance. Their study depicted in Figure 5, emphasized the potential of multi-segmented foam designs to optimize heat transfer via numerical analysis. By improving the transient thermal performance of PCM-based heat sinks, they showed that adjusting the foam filling height ratio could enhance heat transfer during melting under constant heat input.

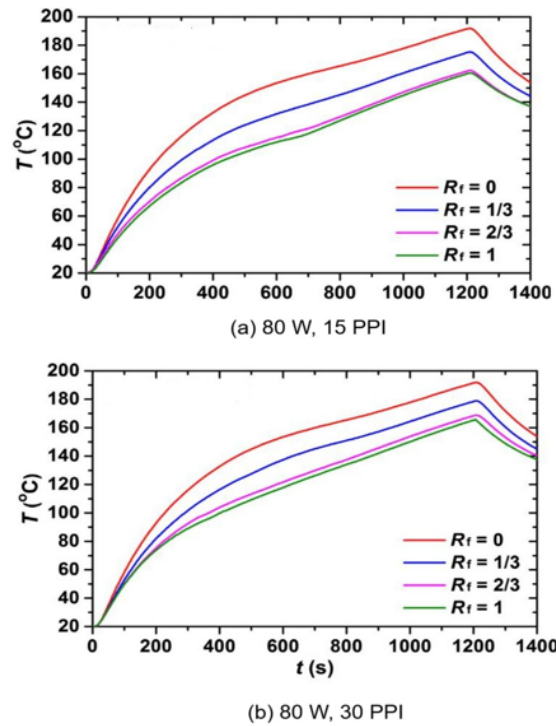


Figure 5. Transient temperature rises during the heating phase of the phase change material-based (PCM-based) heat sink with a partially filled copper foam (filling height ratio (R_f) 0,1/3,2/3,1) having a pore size of (a) 15 pore per inch (PPI) and (b) 30 PPI, adapted with permission from Ref. [50] © 2018 Elsevier

In another investigation, Mahdi and Nsofor [51] applied multiple MF segments in a shell-and-tube PCM TES system. Their findings demonstrated enhanced melting performance due to improved conduction, with system-level geometric effects playing a pivotal role. The study also observed that varying porosity along the heat-flow direction resulted in 7.9% improvement in thermal efficiency and a reduction in melting time. This highlights the significant impact of foam configuration on thermal behavior. Walker et al. [47] also found that segmented MFs with varying porosities improved temperature distribution within the system. These findings contribute to the broader understanding of how modifications to segmentation and porosity can optimize thermal management in PCM systems. Moreover, Buonomo et al. [52] provided a comprehensive review of PCM-enhanced MF TES systems. The review categorized enhancement strategies based on foam characteristics, heat transfer mechanisms, and application scales. Although the use of MFs enhances heat transfer and reduces melting times, the review emphasized a trade-off between performance gains and reduced PCM volume, necessitating a balanced approach to maximize both heat transfer and storage capacity. Lastly, Ferfera and Madani [53] experimentally evaluated the thermal performance of a heat exchanger incorporating combined PCM-metallic foam. Their analysis, conducted under practical operating conditions, demonstrated that copper foam outperformed nickel foam in boosting thermal conductivity, resulting in a 34.5-fold increase in thermal spreading. These findings underscore the significant role of MF materials selection in enhancing the thermal efficiency of PCM-based systems. The incorporation of MFs into PCM-based TES systems has consistently demonstrated improvements in heat transfer, reduced melting and freezing times, and enhanced overall system efficiency. Figure 6, sourced from Scopus data (February 2024) in the Energy and Engineering fields,

illustrates the increasing volume of research on PCM systems with MFs since 2000 [48]. This reflects a growing interest in TES systems, particularly in studies that discuss local thermal non-equilibrium (LTNE) models and the Lattice-Boltzmann method, which are applied at both the pore and representative elementary volume (REV) scales.

Figures 4–6 illustrate the impact of MF porosity and segmentation on the melting behavior of PCM under conduction-dominated conditions. Figure 4 demonstrates how varying MF porosity influences the melting time; as porosity decreases, the melting time is significantly reduced. Figure 5 presents the rise in PCM temperature during the melting process under different heat fluxes (80 W) and porosity values (15 PPI and 30 PPI). This figure compares different foam filling ratios (0,1/3,2/3,1) and examines how both porosity and foam filling ratio affect thermal conductivity and melting time. The experimental conditions employed in this study include constant heating power and standard environmental conditions, with particular emphasis on understanding the effects of foam structure on the PCM thermal performance. Finally, Figure 6 underscores the substantial enhancement in thermal conductivity achieved by integrating MFs into PCMs, highlighting their significant impact on the materials' thermal properties. The data presented in this figure are derived from publications spanning 2000 to 2024, reflecting the increasing research interest in optimizing PCM performance for industrial applications.

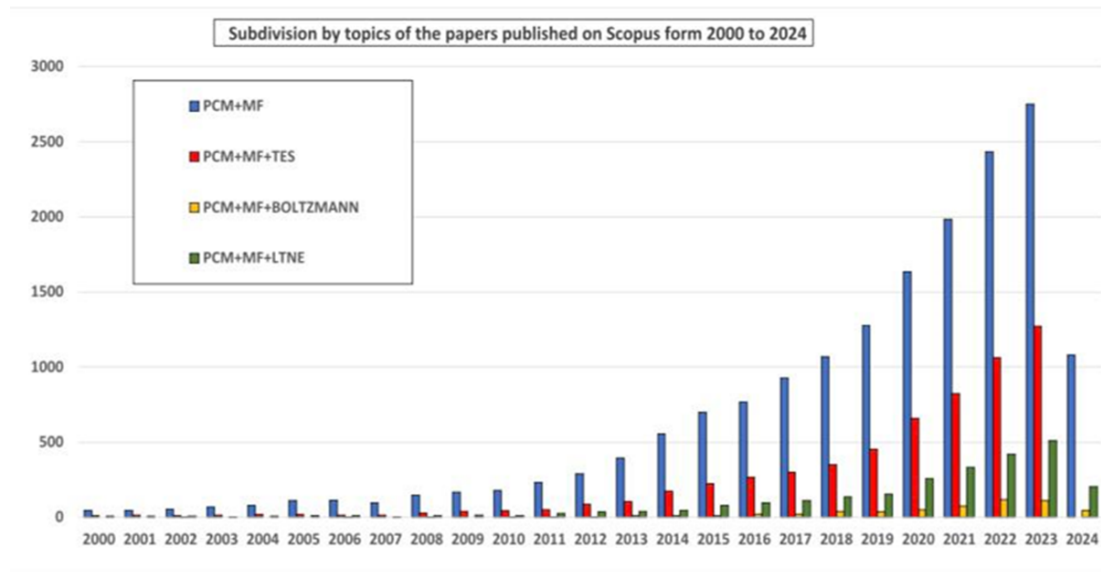


Figure 6. Number of papers published on phase change material (PCM) and metal foam (MF) over the years, adapted from Ref. [52] under the Creative Commons CC BY 4.0 license

4.2 Integration with Nanoparticles

Nano-enhanced PCMs generally improve heat transfer by significantly increasing thermal conductivity through the incorporation of NPs such as multi-walled carbon nanotubes (MWCNT), aluminum oxide (Al_2O_3), copper oxide (CuO), and graphene nanoplatelets (GNP). These NPs not only improve the thermal properties of PCMs but also stabilize the material, preventing phase separation during repeated melting and solidification cycles. The effectiveness of NPs in enhancing PCM performance is influenced by factors such as particle type, concentration, and enclosure configuration. Typically, performance enhancements are defined relative to a pure PCM system under identical operating conditions. The addition of NPs to PCMs creates additional thermal pathways, thereby improving heat transfer and stabilizing the PCM. However, the concentrations of NPs play a critical role in determining overall performance. Higher concentrations may increase the system's viscosity, which can suppress natural convection and consequently reduce overall heat transfer efficiency. For instance, an experimental study investigating the addition of 5% Al_2O_3 NPs to latent heat TES systems found a reduction in melting time from 13 to 5 minutes and a 65% increase in thermal conductivity. This study focused on conduction-based heat transfer and demonstrated that the inclusion of NPs significantly enhanced heat transfer performance relative to pure PCM under controlled heating conditions [54]. In contrast, Karaağaç [55] conducted an experimental evaluation of nano-enhanced PCMs for TES, in which NP dispersion improved thermal conductivity and melting performance. However, the increase in thermal conductivity was only 13%. The study focused on conduction-based heat transfer, with the addition of NPs improving the melting rate under laboratory conditions. Furthermore, the study numerically examined heat transfer enhancement in nano-enhanced PCM, considering both NP addition and enclosure orientation. Using ANSYS Fluent on RT42, the analysis revealed that adding 3% MWCNT or 3% Al_2O_3 and 1% CuO significantly increased the average melting rate. Among these, 3% MWCNT achieved a 3.4% improvement over pure PCM [56]. Additionally, a numerical investigation

examined heat transfer in nano-enhanced PCMs contained in enclosures of various shapes: elliptical, rectangular, cylindrical, and square. The study focused on the interaction between enclosure geometry and the thermal properties enhanced by NP. Simulations with and 4% NPs concentrations (CuO, Al₂O₃) reported a 9.8% improvement in melting and freezing performance, with rectangular enclosures demonstrating a 43% faster thermal response. This suggests that enclosure shape plays a crucial role in optimizing the thermal behavior of nano-enhanced PCM. The study also identified SP24 PCM was optimal for summer conditions, while SP11 was better suited for winter [57]. Experimental work on PLUSICE PCM mixed with 0.1–1% GNP, both with and without SDBS dispersant, reported notable improvements in thermal conductivity. At a 1% concentration, the thermal conductivity increased by 48.83% without SDBS and by 122.26% with SDBS. Additionally, the chemical stability of the materials was maintained up to 257 °C over 1000 cycles, with a 84.95% reduction in light transmittance for solar applications [58]. Mishra et al. [59] investigated the enhancement of thermal conductivity in PCM composites containing carbon black NPs (CBNP). The study highlighted the role of NP concentration in improving thermal performance for energy storage applications. At a 5% CBNP concentration, thermal conductivity increased by 50%. These findings further illustrate that NPs can significantly enhance PCM performance. However, excessive NP concentrations can increase viscosity, thereby impeding heat transfer and highlighting the need for optimized dispersion and composition in achieving maximum performance.

4.3 Integration with Fins

The integration of fins represents one of the most widely adopted strategies for improving the thermal performance of PCM systems. Compared with configurations without fins under identical operating conditions, finned structures significantly enhance heat transfer by increasing the effective heat-transfer surface area and strengthening both conduction and natural convection. As a result, melting rates accelerate, and the system's overall thermal efficiency improves. Experimental studies on PCM melting in a rectangular enclosure have demonstrated that fins accelerate melting by increasing the heat-transfer area and facilitating heat distribution within the PCM domain [60]. Numerical investigations further indicate that fin geometry plays a critical role in determining the heat transfer performance. For example, stepped fins were shown to increase melting rates by 56.3% at 800 seconds and 65.5% at 3600 seconds relative to pure PCM under identical conditions. These improvements were attributed to enhanced heat distribution and altered flow behavior during the melting process [61]. Alternative fin geometries have also been explored to optimize thermal performance. Circular fin configurations, for instance, improved cooling performance by 12.4% and enhanced temperature uniformity by 176.1% in thermal management systems that combine PCMs with pin-fin structures. The results demonstrated that integrating PCMs with extended surfaces significantly improves heat dissipation, making this configuration suitable for electronics cooling applications [23]. Similarly, I-shaped fins have been applied in the battery thermal management system, where they reduced the system temperature from 332.73 K to 330.94 K. The study also examined the influence of mechanical vibration on fin-enhanced PCM systems and reported that vibration further improved heat transfer during melting by strengthening both conduction and convection mechanisms [62]. More advanced fin geometries have also been proposed to further improve PCM thermal performance. Fractal, V-shaped, and tree-like fins were found to reduce wall temperature by up to 27.47% and increase energy storage efficiency by nearly 200%. Experimental investigations revealed that these complex fin structures increase the effective heat-transfer area and enhance melting performance under controlled thermal conditions [63]. Similarly, the use of V-shaped rods as internal structures improved heat distribution and strengthened natural convection, thereby accelerating the melting process [64]. In addition to fin geometry, the number and arrangement of fins strongly influence temperature distribution within the PCM domain, as illustrated in Figure 7.

Increasing the number of fins generally improves heat transfer by expanding the conductive pathways and promoting more uniform temperature fields. Overall, fin integration provides a practical solution to overcome the intrinsic low thermal conductivity of PCMs in TES applications. Previous studies have reported that fin enhancements can improve energy storage efficiency by approximately 19–57%. A comprehensive review of PCM heat transfer enhancement techniques, including fins, porous materials, and nano-enhanced PCMs, confirmed that fin structures remain among the most effective passive enhancement methods [65]. Furthermore, optimized fin orientation can significantly influence melting dynamics. Numerical investigations showed that using two fins inclined at 45° reduced the melting time by approximately 43%. These results highlight that fin inclination modifies convection patterns and strongly affects overall melting performance within rectangular enclosures [66], as illustrated in Figure 8. Material selection also plays an important role in fin effectiveness. High-thermal-conductivity materials, such as copper and aluminum, further reduce melting time by up to 41.6%, as shown in Figure 9. Although copper provides superior thermal conductivity, aluminum offers a more favorable cost-to-performance ratio, making it a practical choice for many applications. This conclusion was supported by studies evaluating the influence of fin materials on PCM melting under identical operating conditions [67]. Finally, fin integration has also been explored in air cooling systems and battery thermal management applications, where it has been reported to reduce system

temperature by approximately 3%. A review of lithium-ion battery thermal management strategies emphasizes the importance of passive and hybrid cooling techniques, with fin-enhanced PCM systems representing a promising approach to improving thermal regulation. The study explored the role of PCMs in mitigating temperature rise and improving thermal stability under practical operating conditions [21]. Furthermore, incorporating fins into a bi-fluid photovoltaic thermal (PV/T) system was shown to improve heat dissipation by 28.3%, as demonstrated by a CFD model of a PV/T system integrating perforated fins and PCM. The study highlighted how the fin design and coupled thermal–fluid effects improved the system’s thermal performance [68]. In general, fin-based enhancement improves PCM melting by promoting heat conduction and influencing natural convection, with the performance strongly dependent on fin geometry and material.

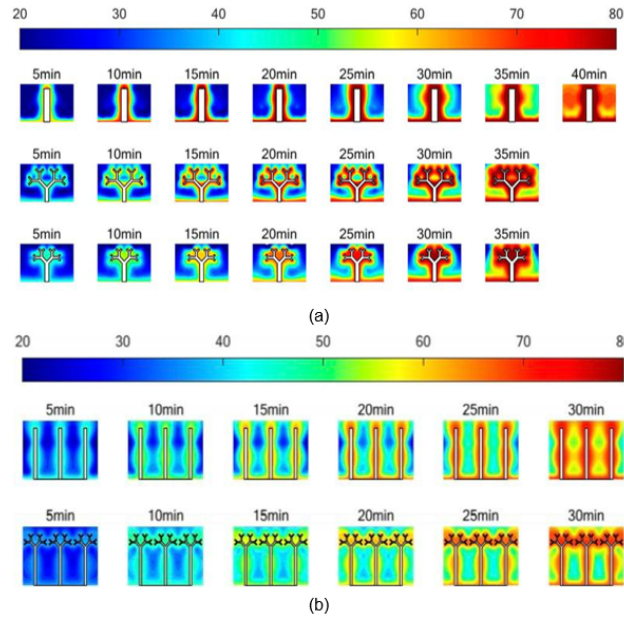


Figure 7. Temperature distribution of units (a) with one fin; (b) with three fins, adapted from Ref. [63] under the Creative Commons CC BY 4.0 license

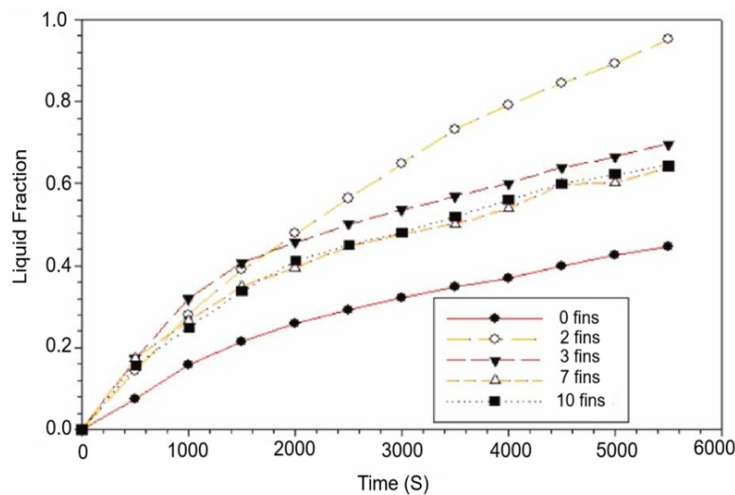


Figure 8. Liquid fraction phenomena of different fin arrangements (0, 2, 3, 7, and 10 fins) as a function of time, adapted with permission from Ref. [66] © 2021 Elsevier

Figures 7–9 illustrate the impact of fin-based enhancement on the thermal performance of PCM systems. These figures illustrate how the number, geometry, orientation, and material of fins affect temperature distribution, liquid-fraction evolution, and the melting rate by modifying heat-conduction paths and natural-convection patterns. Figure 7 presents the temperature evolution at 5-minute intervals, ranging from 5 to 40 minutes for units with one fin, and from 5 to 30 minutes for units with three fins. The color map indicates temperature in degrees Celsius, ranging

from 20 °C to 80 °C. The results demonstrate that adding more fins improves thermal distribution by enhancing heat conduction, leading to a more uniform temperature across the unit. Figure 8 illustrates the progression of the liquid fraction over time during the phase change process for different fin configurations. The data, derived from experimental tests conducted under controlled conditions with a constant heat flux, show that increasing the number of fins enhances heat transfer and accelerates melting. The results indicate that systems with more fins experience faster heat conduction and a higher liquid fraction over time.

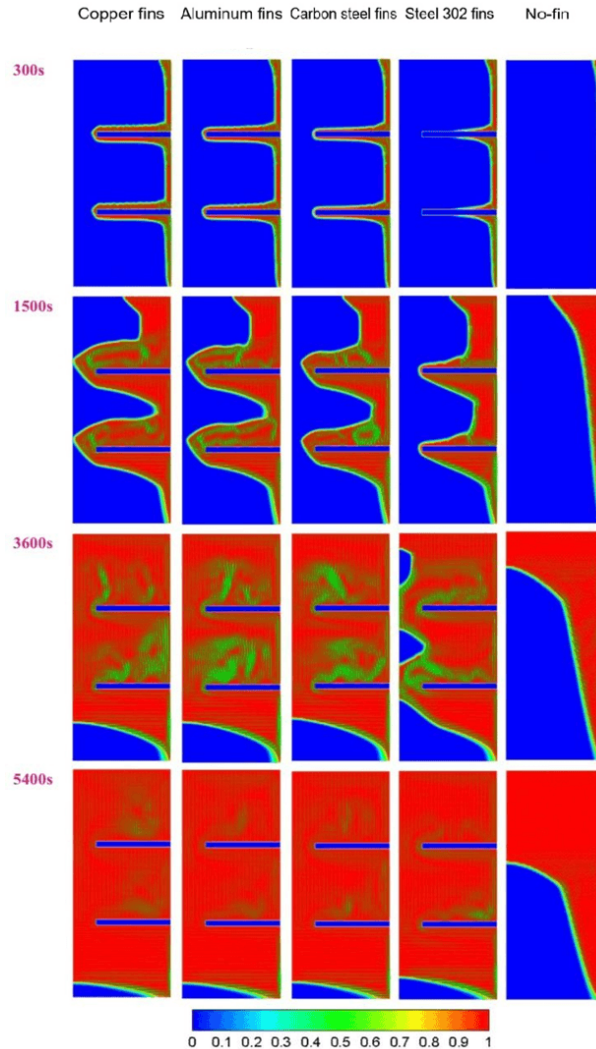


Figure 9. Liquid fraction contours and velocity vectors at different flow times 300 s, 1500 s, 3600 s, and 5400 s for different fin configurations (copper fins, aluminum fins, carbon fins, steel 302 fins, and no fins), adapted with permission from Ref. [67] © 2020 Elsevier

Figure 9 illustrates the temperature distribution and flow dynamics within the PCM during the melting process. The contours represent the progression of the liquid fraction over time, with each column corresponding to a different fin material, while the velocity vectors illustrate the flow behavior of the PCM. The increased fin surface area enhances heat transfer and accelerates melting, with systems utilizing copper and aluminum fins displaying more rapid phase transition. In contrast, systems with 302 fins or no fins at all exhibited slower melting and less uniform heat distribution.

4.4 Integration with Hybrid Techniques

Hybrid systems that integrate MF, NP, and fins have demonstrated the greatest improvements in thermal performance for PCMs. These systems combine the high thermal conductivity of MF, the stabilization provided by NPs, and the increased surface area afforded by fins, resulting in superior overall heat transfer performance. One such hybrid system integrates MF with NPs, such as cyclohexane PCM enhanced with CuO NPs embedded in aluminum MF. This configuration accelerates melting at low porosities 50%, and increases thermal conductivity with higher NP

concentrations. This system achieves a reduction in melting time of up to 90% when using MF with 85–90% porosity and 10–15% NP concentration compared to pure PCM. A TES system based on NP-enhanced PCM embedded in a porous medium demonstrated that combining NPs with porous structures improves heat transfer during melting by enhancing conduction and influencing natural convection within the enclosure, as shown in Figure 10 [69]. The experimental results reported heat transfer rate enhancements of 13% with NPs, 17% with MF, and 24% when both were combined. In contrast, solidification improvements reached 65% with both methods, highlighting that MF had a superior effect over NPs alone. In another study, the effects of NPs and MFs on heat transfer enhancement in PCM were compared. Both techniques improved melting performance, but MFs generally provided stronger conduction enhancement under identical operating conditions [70]. Similarly, in Lauric acid PCM, copper MF combined with Al_2O_3 NPs accelerated melting at low porosities (0.88–0.93). However, at high porosity (0.98), natural convection generated thermal vortices, which slowed the melting rate. Additionally, a numerical study of a rectangular enclosure containing either MFs or NPs found that increasing NP concentration raised viscosity, thereby reducing convective heat transfer. This study emphasized the interaction between conduction enhancement and natural convection in influencing melting behavior [71]. Hybrid systems incorporating 3% NPs and 60% copper MF increased thermal conductivity by 37.7% at an optimal specific surface area of 1600. This was examined in a numerical study that analyzed heat transfer enhancement in PCM by impregnating copper foams with hybrid NPs. The results revealed that the combined approach further improved effective thermal conductivity and melting performance compared with single enhancement methods [72].

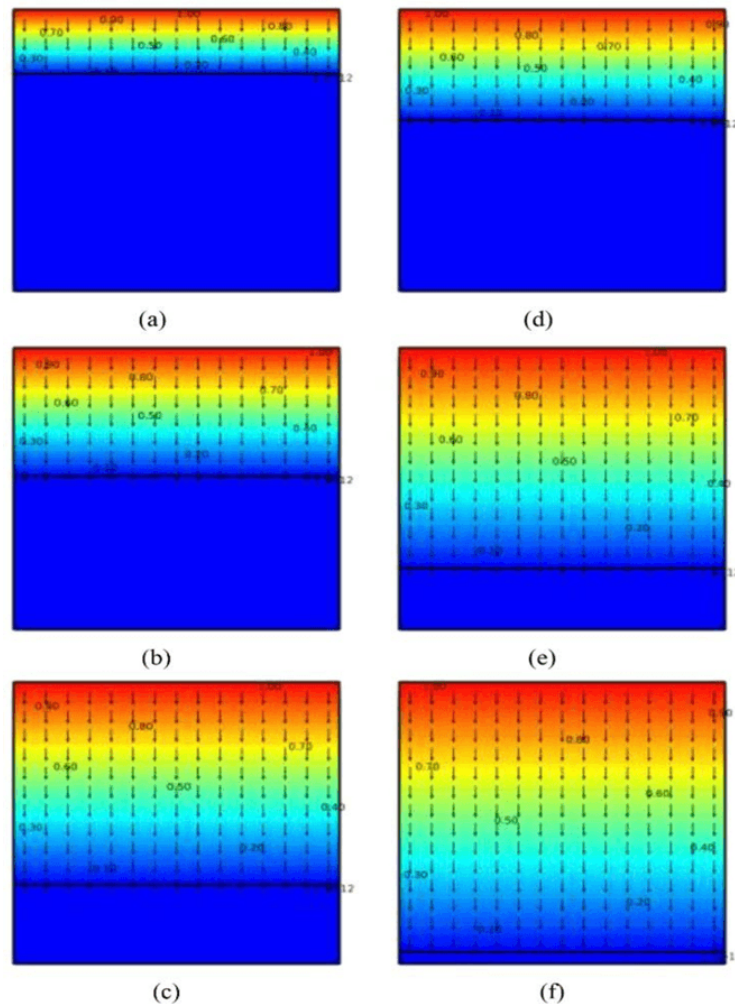


Figure 10. Thermal field (temperature contours and surfaces), heat flux vector, and the location of the melting front at three different times (500 s, 2000 s, and 3000 s), with varying porosity ε (0.95 and 0.85) when the NP volume fraction, ϕ , is 10%

Note: (a) $\varepsilon = 0.95$, $t = 500$ s. (b) $\varepsilon = 0.95$, $t = 2000$ s. (c) $\varepsilon = 0.95$, $t = 3000$ s. (d) $\varepsilon = 0.85$, $t = 500$ s. (e) $\varepsilon = 0.85$, $t = 2000$ s. (f) $\varepsilon = 0.85$, $t = 3000$ s, adapted with permission from Ref. [69] © 2015 Elsevier

The following figures present representative results of the hybrid enhancement technique that combines MFs,

NPs, and fins. These figures illustrate the synergistic effects of combined enhancement methods on temperature fields, melting front evolution, and overall heat transfer performance under different porosity and NP concentration conditions. Figure 10 depicts the temperature distribution within the PCM at various stages of melting, along with heat flux vectors indicating heat flow within the system. The results show that increasing the NP concentration and adjusting the porosity significantly influence the melting rate and thermal conductivity of the PCM. Figure 11 illustrates the temperature distribution, liquid fraction, and the Bejan number, which characterizes the fluid flow behavior and heat transfer efficiency in the system. The results demonstrate that increasing the NP concentration improves thermal conductivity, as evidenced by higher-temperature regions within the PCM. Furthermore, the liquid fraction increases more rapidly at higher NP concentrations, indicating a faster phase change. The Bejan number reflects heat-flow efficiency, with higher values corresponding to better thermal performance.

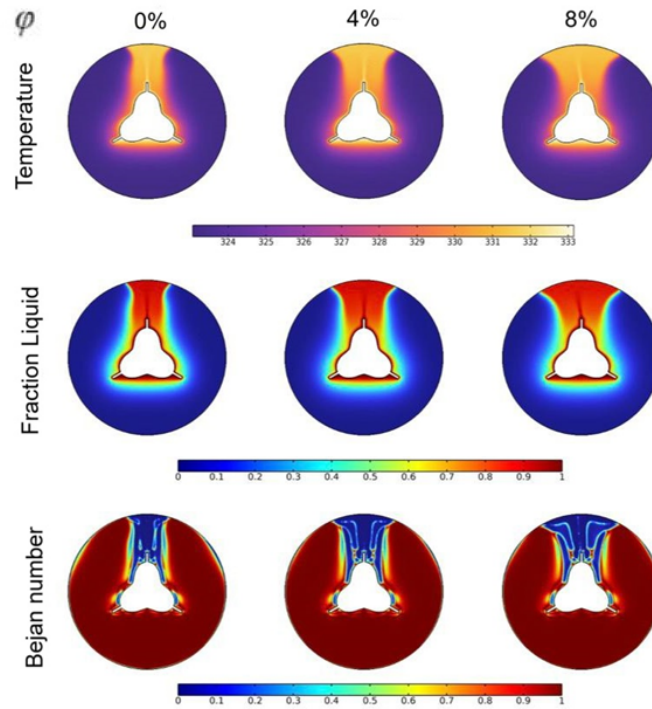


Figure 11. Temperature contours, liquid fraction, and Bejan number for different nanoparticle concentrations (0%, 4%, and 8%) after 20 minutes of thermal exposure, adapted from Ref. [76] under the Creative Commons CC BY 4.0 license

The integration of fins with MF further improves heat transfer in PCM systems. Studies have reported that copper fin–MF composites reduce solidification time by 28.35%, and using 16 fins increased heat transfer by 16.67%. These findings were observed in experiments investigating the solidification process of a fluid-saturated fin–foam composite structure for cold thermal storage. The inclusion of both fins and MF accelerated heat removal and improved solidification performance [73]. Additionally, the integration of aluminum fins and a copper MF in lauric acid-filled heat sinks resulted in a 41.3% improvement in heat transfer rate at optimal porosity (0.9). Higher pore densities further enhanced the thermal response, as demonstrated by the thermal performance analysis of a fined MF heat sink integrated with PCM. This study indicated that the hybrid fin–foam structure enhanced heat dissipation and transient thermal response [74]. In systems using paraffin wax with copper foam fin hybrids, the hybrid structure improved melting time by 52.69–60.52%, depending on the number of fins. As the number of fins increased, thermal efficiency improved by 36.52%, with three fins providing the optimal balance between thermal efficiency and energy storage. A numerical investigation of heat transfer in PCM systems combining MF and fin structures inside an inclined enclosure demonstrated that the hybrid configuration improved melting behavior. The analysis also employed an artificial neural network to evaluate the influence of key design parameters on performance [75].

The combination of fins with NPs has demonstrated a significant synergistic effect on the thermal performance of PCM systems. Studies using modified shell-and-tube PCM systems have reported substantial improvement in melting time when combining fins with NPs. Specifically, at an 8% NP concentration, the melting time decreased by 45% with longer fins, 27% with short fins, and 50% at a 45° tube inclination, as illustrated in Figure 11. These results highlight how geometric modifications, such as fin length and tube inclination, improve heat transfer by increasing the heat transfer area and enhancing conduction [76]. Compared with the T-shaped fin with longitudinal Fe_3O_4 NPs

in TES systems, the study reported improved melting rates due to enhanced thermal conductivity, with performance heavily influenced by NP loading and operating conditions. In this case, the melting time decreases by 34.5% [77]. In addition, the integration of branch-structured fins with Al₂O₃ NPs resulted in an 83% increase in solidification performance when fins were used alone, and an 85% improvement when combined with NPs. In contrast, the use of NPs alone provided only an 8.5–10.3% improvement in the solidification performance of PCM enhanced by branch-structured fins. These results demonstrated that the hybrid approach of combining fins with NPs was far more effective at improving heat transfer than either method used individually [78]. The optimized combination of arc fins and NPs achieves a remarkable 87.03% reduction in melting time. This study, which reviewed PCM-based thermal storage systems for building free-cooling applications, discussed the suitability and limitations of various enhancement techniques in building-scale applications [79].

Collectively, hybrid systems that integrate MF, NP, and fins can enhance thermal conductivity by up to 200% and reduce melting and freezing times by 40–56.5% compared to pure PCM under identical geometry and boundary conditions. While hybrid enhancement techniques offer the highest thermal performance, they also involve higher material costs and greater system complexity than single enhancement methods, which may limit their widespread adoption. From a cost–performance perspective, fin-based enhancement offers a straightforward, cost-effective solution that provides moderate improvements in heat transfer. MFs offer superior thermal performance but at higher material costs and with reduced PCM volume. NP-based enhancement generally offers moderate performance gains while reducing structural complexity. To provide a clearer comparison of the various methods for improving PCM performance, Table 1 summarizes the impact of different enhancement techniques on thermal conductivity, heat distribution, and melting times. Figure 12 complements this by illustrating the relative improvements, thereby helping to identify the most effective approaches for different applications.

Table 1. Comparison of phase change material (PCM) enhancement techniques

Enhancement Method	Enhancing Thermal Conductivity	Heat Transfer Performance Improvement	Melting/Solidification Time Reduction	Benefits	Drawbacks	References
MF	100–200%	30–90% depends on design and foam	66–84.9% reduction	Improved thermal conductivity, better temperature distribution, and increased storage efficiency	Reduces the adequate PCM volume, resulting in higher costs	[40, 45, 52, 80]
NP	48.83–122.26%	3.4–9.8%	3–10% faster	Enhances thermal conductivity, stabilizes PCM, and improves performance in different geometries	Poor dispersion reduces effectiveness, and the high viscosity at high concentrations and the high cost are additional drawbacks	[56–58]
Fins	-	27.47–200%	41.6–65.5% faster	Cost-effective, easy to implement, flexible design (angle shape)	Industrial complexity added weight in some designs	[61, 63, 64, 67],
MF + NP	12–37.7%	12–90%	65–90% reduction	Strong synergy between MF and NP significantly enhances heat transfer	Increased viscosity at high NP, High production costs, and fabrication complexity	[69, 70, 72]
MF + Fins	36.52%	41.3–60.52% reduction	28.35–60.52%	Combines the high conductivity of metal with fins for better thermal distribution	Higher fabrication cost, assembly complexity	[44, 74, 75]
Fins + NP	50%	45–85% improvement	50–87.03%	Significant enhancement from combining fins and NP	The NP effect alone is smaller, and the added design complexity	[76, 78, 79]

Note: MF = metal foam; NP = nanoparticle.

The results summarized in Table 1 are derived from different experimental and numerical studies. Due to differences in geometry, operating conditions, PCM type (e.g., organic and inorganic materials), and enhancement configurations (e.g., porosity, fin geometry, or NP size), direct comparisons of the results may not be fully valid. Factors such as variations in material properties, differences in experimental setups (e.g., enclosure design, heat source location), and discrepancies in boundary conditions (e.g., temperature and pressure ranges), as well as

different numerical assumptions, can significantly influence the performance of PCMs. Therefore, these factors should be carefully considered when comparing results across different studies.

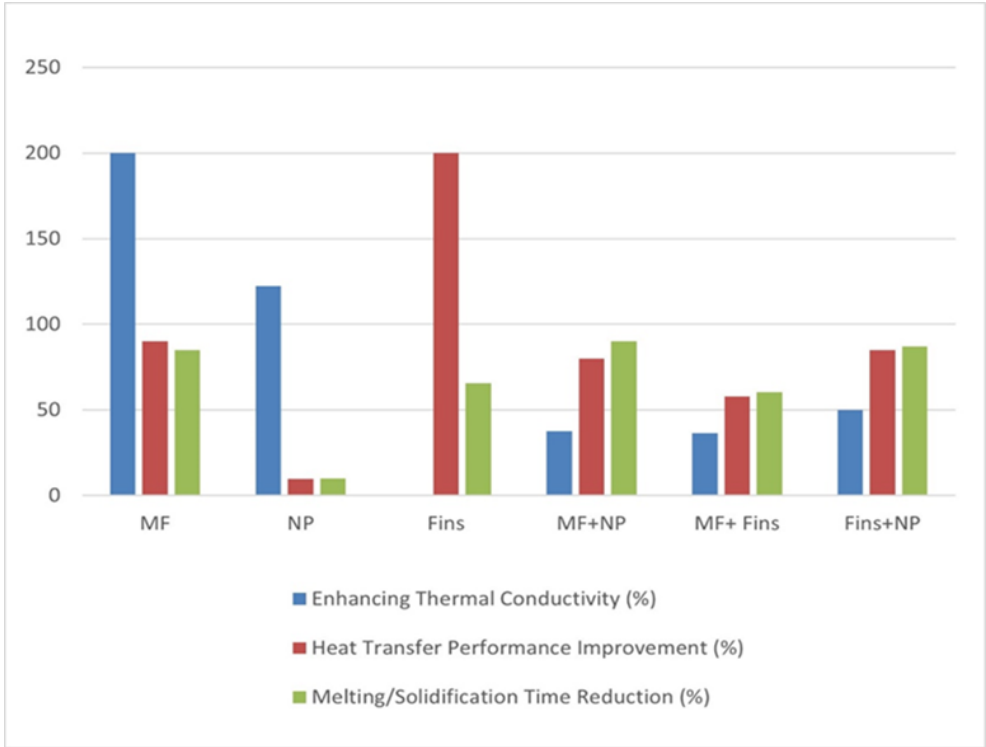


Figure 12. Comparison of phase change material enhancement techniques
 Note: MF = metal foam; NP = nanoparticle.

5 Cost-Benefit Analysis of Improved Techniques

When evaluating enhanced PCMs, it is essential to consider the costs associated with each technique. While integrating MFs significantly improves thermal conductivity and reduces melting times, it also entails higher material and manufacturing costs than simpler methods, such as fins or NPs. Additionally, MFs reduce the available PCM volume for energy storage, a factor that must be considered when determining overall system efficiency. On the other hand, NPs, although effective in enhancing thermal conductivity, can increase viscosity, thereby impeding heat transfer. Therefore, a comprehensive cost-benefit analysis should account for both system costs and actual performance, evaluating the trade-offs between thermal performance improvements and economic feasibility. To strengthen this section, it would be useful to clarify how the cost classification (low, medium, high) was derived. For instance, the classification could be based on reported material prices in the literature, the manufacturing complexity of each technique, and typical fabrication methods used in experimental setups. For instance, “low cost” may refer to methods that use inexpensive materials or simple manufacturing processes, while “high cost” may refer to techniques that require specialized materials or advanced fabrication methods. Providing a brief explanation of these classification criteria, or referencing supporting studies, would add transparency to the cost comparison.

To facilitate a qualitative comparison of various PCM enhancement techniques from a cost-performance perspective, Table 2 presents a relative classification of common methods.

Table 2. Qualitative cost-performance classification of phase change material (PCM) enhancement techniques

Technique	Relative Cost	Performance Gain	Suitable Applications
Fins	Low	Medium	Buildings, HVAC
NPs	Medium	Low-Medium	Compact systems
MF	High	High	EVs
Hybrid	Very High	Very High	High-end TES

Note: MF = metal foam; NP = nanoparticle; EVs = electric vehicles; TES = thermal energy storage; HVAC = heating, ventilation, and air conditioning.

6 Challenges and Future Research Directions

Despite significant advancements in PCM performance, several challenges remain for broader industrial applications, including:

- High Material Costs

Research direction: The cost of advanced materials such as MFs and NPs remains a barrier to large-scale implementation. Future research should focus on developing cost-effective alternatives without compromising performance.

- Manufacturing Complexity

Research direction: The intricate manufacturing processes required to integrate enhanced materials into PCMs hinder mass production. More efficient and scalable manufacturing methods need to be developed to make these technologies commercially viable.

- Compatibility with Emerging Technologies

Research direction: The integration of PCMs with technologies, such as thermoelectric devices and heat pumps, shows significant promise. Future research should optimize PCMs for compatibility with these systems to improve overall performance and efficiency.

- Sustainable Materials

Research direction: The development of environmentally friendly, bio-renewable materials that enhance thermal efficiency is critical to the long-term adoption of PCMs in energy systems.

- Manufacturing Optimization

Research direction: Future studies should focus on simplifying the production processes of composite PCMs to make them more cost-effective and scalable, improving commercial viability.

- Integration with Emerging Technologies

Integrating PCMs with technologies like thermoelectric devices and heat pumps can improve energy storage and conversion efficiency. Future research should focus on optimizing this integration to expand the potential applications of PCMs, particularly in sectors like renewable energy and waste heat recovery.

7 Implications for Energy Systems and Management

The performance enhancement of PCMs using fins, MFs, NPs, and hybrid configurations directly impacts various energy systems. The choice of enhancement technique is influenced by factors such as system scale, thermal response requirements, cost constraints, and operational needs.

- Fin-based Enhancement

This technique is ideal for building-scale applications where moderate performance improvement, low cost, and ease of manufacturing are needed. In buildings, fin-based PCMs help improve energy efficiency by reducing the demand for heating and cooling systems. These systems are generally small to medium-sized, and the operational focus is on balancing thermal storage and thermal comfort.

- MF-enhanced PCMs

MF-enhanced PCMs are suited for compact systems that require rapid heat charging and discharging, such as in battery thermal management and concentrated solar power (CSP) systems. While material costs are higher, the improved heat-transfer performance justifies them, especially in EVs and solar applications. The system size tends to be small to medium, depending on the specific application, and operational challenges include thermal cycling stability.

- NP-enhanced PCMs

NP-enhanced PCMs offer moderate performance improvements and are easily integrated into systems. However, their long-term stability and potential viscosity-related effects should be considered in applications requiring consistent thermal performance. These materials are suitable for electronic cooling or solar water heaters, where thermal demands are lower. The system size is typically small to medium, and the main operational consideration is ensuring material longevity.

- Hybrid Systems

Hybrid systems, combining fins, MFs, and NPs, are well-suited for advanced TES and waste-heat recovery systems, where higher performance justifies the increased complexity and cost. These systems are commonly used in industrial applications, where large-scale energy storage is necessary. The key operational factors are scalability and long-term stability, ensuring reliable performance.

- Hybrid Enhancement Techniques

Hybrid techniques deliver the highest performance and are best suited to advanced TES and waste-heat recovery systems, where performance is a priority despite the added complexity and cost. These systems are often used in large-scale energy storage and heat recovery applications, where high thermal energy density is essential. Operational challenges focus on optimizing system integration and ensuring cost-effectiveness while maintaining high performance.

- Practical Deployment Considerations

For practical deployment, scalability, long-term cycling stability, and system integration must be prioritized to ensure reliable operation in real-world energy production. Research should focus on developing cost-effective manufacturing processes while maintaining high thermal performance. Overcoming these challenges will support the widespread adoption of PCM technologies across sectors such as renewable energy storage, EVs, and industrial waste heat recovery.

8 Conclusion

Improving the performance of PCMs requires a balanced integration of different enhancement techniques, considering both thermal performance and cost-effectiveness. Hybrid systems combining MF, NPs, and fins offer the most significant overall benefits. While MF provides the most substantial improvement in heat transfer, the high materials costs and manufacturing complexities can limit its widespread adoption. NPs and fins, though more cost-effective, provide moderate performance gains. MF can increase thermal conductivity by up to 200% and reduce melting time by up to 80% compared to pure PCM. However, these improvements depend on the type of MF material, porosity, and PPI characteristics. Despite the thermal benefits, the high material costs and manufacturing challenges restrict its application, particularly in cost-sensitive industries. Based on the reviewed studies, MFs remain the most effective single technique for accelerating both melting and solidification. NPs provide notable improvements in thermal conductivity, with a maximum increase of 65% at a 5% Al_2O_3 concentration relative to the pure PCM. However, their concentration must be carefully controlled to avoid viscosity issues, which could impede heat transfer. Fins have been reported to reduce melting time by 65.5% compared to pure PCM, making them the most practical and cost-efficient option when budget constraints and manufacturing simplicity are critical. From a design perspective, fin-based enhancement is well-suited to low-cost, compact systems with moderate thermal response. Conversely, MFs are better suited for applications requiring rapid melting and high heat-transfer rates. NPs-enhanced PCMs offer moderate improvement with easier integration, while hybrid techniques are ideal for high-performance systems where increased complexity and cost are justified. Future research should focus on reducing material costs, simplifying manufacturing processes, and optimizing hybrid configurations to enhance both performance and storage capacity without introducing additional trade-offs. Furthermore, most existing studies are limited to small laboratory-scale systems, with insufficient emphasis on large-scale performance and long-term stability. Additional efforts are needed to standardize testing conditions, conduct extended thermal cycling, and validate LTNE-based numerical models to facilitate the transition of PCM enhancement techniques from research to reliable industrial applications.

Author Contributions

Conceptualization, B.S.Y. and S.L.G.; methodology, B.S.Y. and S.L.G.; formal analysis, B.S.Y.; investigation, B.S.Y.; data curation, B.S.Y.; writing—original draft preparation, B.S.Y.; writing—review and editing, S.L.G.; resources, B.S.Y. and S.L.G.; supervision, S.L.G.; validation, S.L.G. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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Nomenclature

PCMs	Phase change materials
TES	Thermal energy storage
MFs	Metal foams
NPs	Nanoparticles
HVAC	Heating, ventilation, and air conditioning
EVs	Electric vehicles
PPI	Pore per inch
HTF	Heat transfer fluid
LTNE	Local thermal non-equilibrium
MWCNT	Multi-walled carbon nanotubes
Al ₂ O ₃	Aluminum oxide
CuO	Copper oxide
GNP	Graphene nanoplatelets